

PERSONAL-SAFETY PLATFORM · COMPLETE APP GUIDE

HearMe.

A personal-safety mobile app paired with the NeuroBand wearable — designed to detect distress, capture evidence, and reach trusted contacts when seconds matter.



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1 · What HearMe is

HearMe is an integrated personal-safety toolkit. The phone app is the user-facing surface; the optional **NeuroBand** wristband adds physiological detection of duress that doesn't depend on the user being able to reach for a phone. Supabase provides authenticated, row-level-secured cloud sync for evidence, and an opt-in dashboard surface lets vetted partner organisations help dispatch responders.

The three things HearMe is built to do

Detect

Detect distress automatically — voice triggers, biosignal fusion, behavioural anomalies, fall and crash detection, acoustic events, BLE stalker scanning, and isolation scoring.

Capture

Capture evidence the moment it matters — pre-roll audio, photos, location pings, biosensor frames — all chained, signed, and time-stamped on-chain.

Reach

Reach trusted people fast — SMS to all contacts, ack-aware tier escalation, encrypted live-share, witness mode, community SOS mesh, and PSAP hand-off where regulations allow.

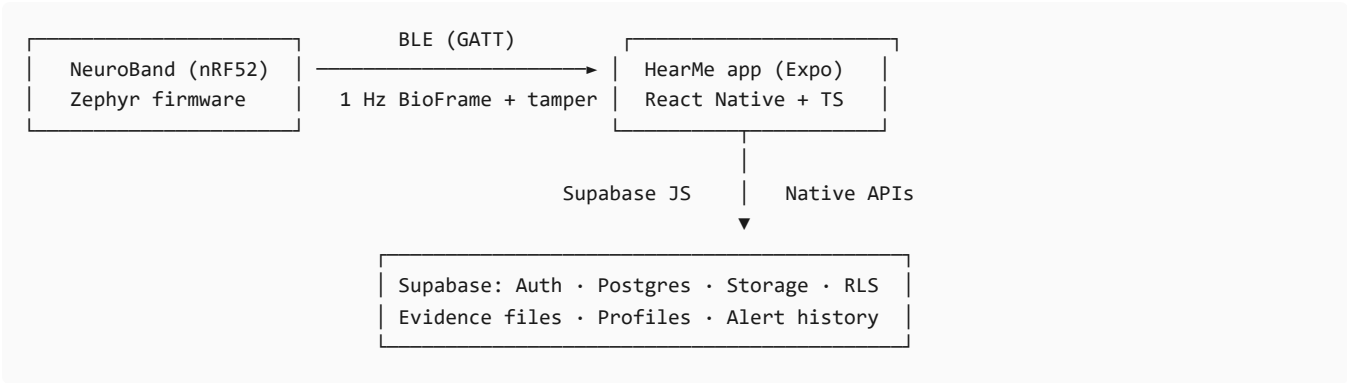
Design principles

- **Phone is the source of truth.** The phone runs detection, the band augments. If the band is down or absent, the app degrades gracefully.
- **Cloud is the destination, not the operator.** Evidence is encrypted on the device, hashed locally, then pushed. Supabase never sees plaintext audio.
- **Court-grade chain of custody.** Every captured artefact is hashed, chained, signed with a device-bound Ed25519 key, and anchored to the Bitcoin blockchain via OpenTimestamps.
- **Hostile-environment design.** Disguise mode, decoy lock, force-unlock detection, anti-uninstall, airplane-mode counter — every adversarial path is closed.
- **Honest where adapters are needed.** When a feature genuinely needs a native module / vendor SDK / hardware (PSAP API, satellite SDK, ML Kit), the JS layer reports "not available in this build" — never returns silent success it didn't earn.

The shape of this guide. Sections 4–7 cover the baseline app shipped at version 1.0. Sections 8–15 cover the 35 advanced modules added in the subsequent rev. Each advanced module includes its purpose, why it exists, and the file path of the implementation.

2 · High-level architecture

Three actors: the band, the phone, and the cloud. Detection happens on the phone, augmented by the band. Cloud sync happens only when an event fires.



Layer map

Layer	Lives in	Responsibilities
UI / routes	app/ (expo-router)	File-based navigation; one file per screen.
Reusable widgets	components/	SOSButton, GlassCard, themed primitives.
Cross-tree state	providers/	Auth · Theme · Language · Voice · Accessibility.
Domain logic	lib/	All non-UI code: BLE, fusion, evidence, relays, adapters.
Schema & RLS	supabase/	SQL migrations; tables, policies, RPCs.
Firmware	firmware/neuroband/	Zephyr app for the nRF52840-DK.

Data residency

- **On-device only:** raw audio, draft evidence sessions, voiceprint, signing keys, duress PIN, cycle-log, voice-trigger challenges.
- **On-device, then encrypted to cloud:** finished evidence sessions (file content + chain hash + signature + OTS receipt).
- **Cloud-only ephemeral:** live-share ciphertexts (E2E AES-GCM), guardian-relay ack messages, community-mesh beacons (auto-prune on TTL).
- **Public RPCs:** incident heatmap (k-anonymous aggregate, never per-user), evidence_attestation lookup (chain hash + signature + OTS receipt only).

3 · Technology stack

Layer	Choice	Why
App runtime	Expo SDK 54, React Native 0.81, React 19	One codebase, two stores, fast iteration, native modules when needed.
Routing	<code>expo-router</code>	File-based — every screen is a file under <code>app/</code> .
Language	TypeScript 5.9	Compile-time guarantees on every boundary.
State	React Context	Auth, Theme, Language, Voice, Accessibility — no global store needed.
Backend	Supabase (Postgres + Auth + Storage + RLS)	Per-row security; phone OTP; private buckets with signed URLs.
BLE	<code>react-native-ble-plx</code>	Background-enabled; mature; supports peripheral writes for haptic codes.
Sensors	<code>expo-sensors</code> , <code>expo-location</code> , <code>expo-camera</code>	Accel · Gyro · LightSensor · GPS · camera in one toolkit.
Crypto	<code>expo-crypto</code> · <code>@noble/ed25519</code> · <code>@noble/ciphers</code>	Hashing + RNG via Expo; pure-JS audited Ed25519 + AES-GCM.
Auth	Supabase OTP, biometrics, Aadhaar, demo PIN	Multiple paths; works without phone signal.
i18n	<code>i18next</code> + <code>react-i18next</code>	Multi-language support including spoken navigation.
Firmware	Zephyr / nRF Connect SDK on nRF52840	BLE-first low-power MCU; standardised GATT.

4 · Baseline features — emergency response

① SOS Button

SHIPPED

One-tap silent alarm that fires SMS with live location to all trusted contacts. Optional auto-dial of the configured emergency number after the SMS pipeline completes.

`app/(main)/(tabs)/index.tsx · lib/emergency-sms.ts`

② Voice Trigger

SHIPPED

Listens for distress phrases or sustained loud audio and auto-fires SOS. Honest about what it can do without a native KWS module: amplitude-VAD detection of screams / shouted safe words. The native module slot lets a Whisper / Vosk / Porcupine backend drop in.

`lib/voice-trigger.ts`

③ Triple-tap & Duress PIN

SHIPPED

Discreet activation paths when the screen can't be seen. The duress PIN is the user's normal PIN reversed — typing it unlocks normally to an attacker but silently fires SOS. An explicit duress PIN is also supported for users whose normal PIN is a palindrome.

`lib/duress.ts · app/lock.tsx`

④ Fake Call

SHIPPED

Simulates an incoming call to defuse a hostile situation. Configurable delay; UI matches platform-native call screen.

`app/(main)/fake-call.tsx`

⑤ Siren

SHIPPED

Loud alarm to deter attackers and attract attention. Plays through the phone's speaker and (if registered) the smart-home siren module fans the alarm out to Hue lights and Google/Alexa speakers.

`lib/siren.ts`

⑥ Helplines

SHIPPED

Quick-dial directory: local emergency services and women's helplines. Country-aware where possible.

`app/(main)/helplines.tsx`

5 · Baseline features — evidence & monitoring

⑦ Audio Recorder

SHIPPED

Encrypted recording uploaded to the evidence locker. Metering enabled so callers can also extract envelope data. Recordings persist on device until uploaded; on-device list, playback, and delete.

lib/audio-recorder.ts · app/(main)/audio-recorder.tsx

⑧ Evidence Locker

SHIPPED

Supabase-backed cloud storage for audio, photos, and incident logs. Each item carries a content-hash; items are linked into a chain ($\text{chain}[i] = \text{SHA256}(\text{chain}[i-1] || \text{content}[i])$) so any tampering invalidates every later hash. The locker is private — files are accessible only via signed URLs scoped to the owning user.

lib/evidence-locker.ts · lib/evidence-cloud.ts · supabase/evidence.sql

⑨ Hidden-camera Detector

SHIPPED

Uses the phone's camera to spot infrared LED reflections — common in covert surveillance cameras. Live-view scan with grid overlay so the user can sweep a room.

app/(main)/camera-detector.tsx

⑩ Journey Monitor

SHIPPED

Tracks a route to a destination and alerts contacts on deviation, no-show, or check-in failure. Background-aware where the platform allows. Driver/passenger classification (advanced module #12) softens false alarms for passengers.

lib/journey-monitor.ts · app/(main)/journey-monitor.tsx

⑪ Check-in Timer

SHIPPED

Auto-alerts contacts if you don't confirm safety in time. Inspired by the original Kitestring concept — single active check-in stored in settings, expiry epoch, dismiss button.

lib/timer-checkin.ts · app/(main)/check-in.tsx

⑫ Behaviour Monitor (BBA)

SHIPPED

Behavioural Biometric Authentication. Continuously scores the active session using the on-device fraud model and prompts for PIN / biometric re-verification when interaction patterns look unusual. Originally derived from the SudoCode BBA model.

lib/bba-model.ts · lib/behavior-tracker.ts · lib/behavior-detector.ts · app/(main)/behavior-monitor.tsx

⑬ Alert History

SHIPPED

Chronological log of every triggered event — manual SOS, fusion fire, journey deviation, check-in lapse, self-test pass / fail. Filterable; supports full export.

lib/alert-history.ts · app/(main)/alert-history.tsx

6 · Baseline features — privacy, accessibility, identity

14 PIN, Biometric, Aadhaar

SHIPPED

Multi-path identity. Demo PIN works without backend; biometric uses platform Face ID / fingerprint; Aadhaar OTP-based path supports India e-KYC. Each path is independent — losing one does not lock the user out.

`lib/aadhaar.ts · lib/otp.ts · lib/demo-auth.ts · app/aadhaar.tsx · app/login.tsx · app/verify-otp.tsx`

15 Disguise Mode

SHIPPED

The app masquerades as a calculator on the home screen. Hidden swipe / tap pattern reveals the real UI. Distinct from the decoy-lock module (advanced #15) which fakes a lock screen *after* the app is open.

`app/disguise.tsx`

16 App Lock

SHIPPED

Re-auth gates on sensitive screens (evidence locker, contacts, settings). Force-unlock detection (advanced #16) layers struggle-aware bio guard on top.

`app/lock.tsx`

17 Voice Guide

SHIPPED

Fully spoken navigation for visually-impaired users. Built on i18next (translations) + expo-speech (TTS).

`lib/voice-guide.ts · lib/nav-guide.ts · lib/nav-guide-ref.ts`

18 Multi-language

SHIPPED

i18next + react-i18next. Adding a new locale = adding a JSON file under `lib/i18n/`.

`lib/i18n.ts · app/language.tsx`

19 Themeable UI

SHIPPED

Light, dark, and high-contrast variants. Theme provider exposes tokens; every screen is theme-aware out of the box.

`providers/* · constants/*`

20 Onboarding

SHIPPED

Guided setup that walks a new user through identity, contacts, permissions, and PIN/biometric enrolment. The `onboardingComplete` flag in settings gates the main app routes until done.

`app/onboarding.tsx · app/setup-pin.tsx`

7 · NeuroBand wearable integration

The NeuroBand is an optional Bluetooth wristband running Zephyr firmware on an nRF52840. It exposes a single GATT service over which it streams a 20-byte BioFrame at 1 Hz and emits sparse silent-trigger / tamper events. The phone-side fusion engine combines those signals with phone sensors to detect duress with high specificity.

BioFrame contents

Field	Type	Purpose
seq	uint16	Per-connection rolling counter for ordering.
hr	uint8	Heart rate (BPM).
hrvRmssd	uint8	Heart-rate variability proxy.
spo2	uint8	Blood-oxygen %.
gsrUs	uint16	Skin conductance in μ S.
skinTempC	int8	Skin temperature, fixed-point $^{\circ}$ C.
motion	uint8	still / walk / run / fall / unknown.
semg	uint8	Surface-EMG envelope (clench detection).
battery	uint8	Band battery %.
flags	bits	charging · tamper · low-battery.
counter	uint32	Replay-protection monotonic counter.

Fusion engine

The §8 Volume-III duress signature: any two-of-five physiological signals firing together — while motion is still or walk — fires a verdict. Each marker has a *sustained* requirement (e.g. HR must be elevated for ≥ 4 s) so single-frame sensor noise can't trigger SOS. Sensitivity is adjustable; workout-mode lockout suppresses fires while exercising.

```
five markers → per-marker fired? (instantMarkers)
               |
               ▼
           sustained-window memory (SUSTAIN_REQUIRED per marker)
               |
               ▼
fire = sustainedCount ≥ requiredCount (sensitivity-tuned)
```

Mock mode

A synthetic BioFrame stream lets the entire UI + fusion engine run without paired hardware. Three modes — calm, active, duress — exercise the rule transitions.

[lib/neuroband-ble.ts](#) · [lib/neuroband-fusion.ts](#) · [lib/neuroband-pairing.ts](#) · [lib/neuroband-storage.ts](#) · [firmware/neuroband/](#)

8 · Advanced features — Phase A · evidence foundation

Phase A — Evidence foundation

Make every captured byte court-defensible and recoverable.

1 Continuous pre-roll buffer

SHIPPED

Always keeps the last 30 seconds of audio on disk. When SOS fires, the buffer is attached to the evidence session retroactively — we capture what happened *before* the trigger, not just after. Implementation rotates 10-second WAV chunks; a native circular buffer can drop in via the backend slot.

`lib/preroll-buffer.ts`

2 Cryptographic chain-of-custody

SHIPPED

Every uploaded evidence session is signed with a per-device Ed25519 key (private key in SecureStore, public key uploaded to Supabase) and the chain hash is anchored to the public OpenTimestamps Bitcoin calendar. Verifiers can prove "this evidence existed by block N" without trusting HearMe.

`lib/evidence-signing.ts` · `supabase/evidence-signatures.sql`

3 Dead-man's switch upload

SHIPPED

Persistent retry queue with reachability probing. If primary upload fails, registered relay backends (BLE-paired peers, alternate transports) get a chance. After three failed attempts contacts are SMS-notified that evidence is pending so the device can be physically recovered.

`lib/evidence-deadman.ts`

4 Lawyer / NGO export package

SHIPPED

One-tap signed bundle: `manifest.json` with chain hash, signature, public key, OTS receipt, and signed Supabase URLs for every media item; plus a standalone `summary.html` a legal reader can open offline. The recipient's verification recipe is included inline.

`lib/evidence-export.ts`

9 · Advanced features — Phase B · trusted network

Phase B — Trusted network

Reach the right person, fast — with proof of receipt.

5 Encrypted live-share session

SHIPPED

End-to-end AES-GCM stream of location (and optional audio chunks) over Supabase Realtime. The AES key lives only in the URL fragment after the # — Supabase only ever sees ciphertext + the channel id. Recipients send back ack receipts so the user knows who actually opened the link.

lib/live-share.ts

6 Guardian relay (ack-aware fan-out)

SHIPPED

Contacts have priority tiers. Tier 1 is messaged immediately; if no ack within T seconds, Tier 2 is messaged next, then Tier 3 — and the configured emergency number (e.g. 112) is dialled as the last-resort step. Acks come over a per-alert Realtime channel.

lib/guardian-relay.ts · lib/types.ts

7 Two-way duress

SHIPPED

A trusted contact can send "are you safe?" — the user's band buzzes a covert 3-pulse pattern. The user replies via band triple-tap (safe) or button (unsafe), or silence for 60 s auto-escalates to SOS via the relay.

lib/two-way-duress.ts

8 Witness mode

SHIPPED

The user shows a one-time QR; whoever scans joins an E2E-encrypted session and their phone records audio + location chunks straight into the user's evidence locker — anonymously to the user, signed by the witness's device.

lib/witness-mode.ts

9 Community SOS mesh

SHIPPED

Opt-in: when SOS fires, a short-lived geo-anonymous beacon is published. Other HearMe users within R metres get a non-blocking "someone nearby just sent SOS" banner; they can offer "on the way" with their location. Beacon ids rotate; recipients are anonymous unless they explicitly identify.

lib/community-mesh.ts · supabase/community-mesh.sql

10 · Advanced features — Phase C · detection & sensing

Phase C — Detection & sensing

Notice the threat before the user has to.

10 BLE stalker scan

SHIPPED

Detects covert personal trackers (AirTag, Samsung SmartTag, Tile, Google Find-My) that have followed the user across distinct locations for ≥ 30 minutes. Manufacturer-data parsing handles MAC rotation; persistent sighting log reaches across app sessions.

`lib/stalker-scan.ts`

11 Isolation score

SHIPPED

Combines BLE neighbour count, ambient dBFS, lux (Android), speed, hour-of-day, and safe-zone membership into a 0–100 score. High score auto-raises fusion sensitivity and prompts the user to start journey monitoring.

`lib/isolation-score.ts`

12 Driver / passenger classifier

SHIPPED

Heuristic on speed + accelerometer / gyroscope variance + touch rate, classifying the user as *driver*, *passenger*, *on-foot*, *idle*. The journey monitor uses this to soften false alarms when the user is clearly a passenger.

`lib/drive-passenger.ts`

13 Phone-tamper detection

SHIPPED

Triple-coincidence detector: screen-off without a recent touch + accelerometer shock + simultaneous network loss within 6 s $\Rightarrow$ silent SOS. Two-of-three only $\Rightarrow$ heightened-alert banner.

`lib/phone-tamper.ts`

14 Off-grid mode

ADAPTER + DEFAULT

Network watchdog raises a banner on sustained connectivity loss. Pluggable transport adapters (Meshtastic LoRa, Garmin inReach, sat-phone bridges) drop in via the backend interface; iOS satellite-SOS is surfaced as a help-text prompt because Apple gates the trigger to the side button.

`lib/offgrid.ts`

11 · Advanced features — Phase D · resilience

Phase D — Resilience & adversarial robustness

Stay useful when the attacker has the phone.

15 Decoy lock screen

SHIPPED

Different from disguise mode (calculator) and duress PIN (reversed). The decoy lock is a fake lock screen the user can flip on at any moment: it shows "No Service", a draining fake battery, and rejects every input as "incorrect" — while silently firing SOS in the background.

```
lib/decoy-lock.ts
```

16 Force-unlock detection

SHIPPED

Wraps `expo-local-authentication` with motion-aware verification. Samples accelerometer for 2 s before the bio prompt; if motion variance is high *and* failed bio attempts within window ≥ 2 , the unlock is flagged as forced and silent SOS fires.

```
lib/force-unlock.ts
```

17 Anti-uninstall guard

ADAPTER + RECIPE

JS adapter + a complete Android Device Admin recipe (Manifest + receiver + DPM calls). The duress PIN is the gate to disable. iOS path leans on Screen Time restrictions because Apple does not let third-party apps prevent uninstall.

```
lib/anti-uninstall.ts
```

18 Airplane-mode counter

SHIPPED

Detects simultaneous loss of cellular AND Wi-Fi within seconds of an accelerometer shock — the signature of someone toggling airplane mode after grabbing the phone. Escalates via a registered alt-modem backend (e.g. NeuroBand LTE-M, sat dongle) or the standard relay.

```
lib/airplane-counter.ts
```

12 · Advanced features — Phase E · on-device intelligence

Phase E — AI / on-device intelligence

Smarter detection without sending data to a server.

19 Threat-context classifier

SHIPPED

Multi-modal score combining transcript, fusion verdict, isolation, drive context, acoustic events, time of day, and safe-zone membership. The default is a transparent rule-based heuristic; an on-device LLM (Gemma 2B / Phi-3-mini via mlc-llm) can drop in via the backend interface.

```
lib/threat-classifier.ts
```

20 Acoustic event detection

SHIPPED

Default envelope detector identifies scream, shout, thud, silence from the meter signal. A YAMNet TFLite backend slot accepts the same event shape; the recipe (audio windowing, ontology mapping, native dep list) is bundled in the source.

```
lib/acoustic-events.ts
```

21 Speaker diarization

SHIPPED

Splits an evidence audio clip into "user vs other" segments. Default uses RMS + zero-crossing-rate + spectral centroid against an enrolled voiceprint stored in SecureStore. Adapter slot accepts pyannote / TitaNet / ECAPA-TDNN backends.

```
lib/diarization.ts
```

22 Auto-redaction

ADAPTER + RECIPE

Face / plate / PII blur on captured photos. Refuses to silently return originals — the no-op default reports `{ ok: false, reason: "noBackend" }`. ML Kit (Android) and Vision (iOS) recipes are inline; manual-region blur is supported the same way.

```
lib/auto-redaction.ts
```

23 Deepfake voice-trigger guard

SHIPPED

Wraps voice-trigger with a voiceprint-match step (re-using the diarization enrolment) and an optional per-session challenge word. A failed challenge demotes the response from "act-loud" to "prompt-user" — never silently rejects, because false negatives in safety code are worse than false positives.

```
lib/voice-trigger-guard.ts
```

13 · Advanced features — Phase F · NeuroBand expansion

Phase F — NeuroBand expansion

Make the wearable smarter and more personal.

24 HRV-based stress baseline

SHIPPED

24 circadian-hour buckets, each tracking RMSSD mean/variance via Welford's online algorithm. The current frame's z-score against the right hour bucket gives a 0–100 stress index that fusion uses to dynamically adapt sensitivity.

```
lib/hrv-baseline.ts
```

25 Cycle-aware sensitivity

SHIPPED

The luteal phase shifts resting HR up ~5 BPM and HRV down ~15%, which produces chronic false-positive runs in fixed-threshold fusion. The module ships a privacy-preserving manual log + a HealthKit / Health Connect adapter slot, then emits an adjustment {phase, hrThresholdOffsetBpm, hrvBaselineMultiplier}.

```
lib/cycle-aware.ts
```

26 Band-side LED / haptic codes

SHIPPED

Extended single-byte command protocol: 0x10–0x1F acks, 0x20–0x2F warnings, 0x30–0x3F silent statuses. Each code maps to a distinct vibration pattern + LED colour/pulse so the user feels-through-the-wrist what's happening (SMS sent, contact opened, helper on the way...) without looking at the phone.

```
lib/band-feedback.ts
```

27 Group band roaming

SHIPPED

One account, multiple bands — kid + parent + grandparent. Per-member geofence, group realtime channel for fusion fires from any band, role labels (self / kid / parent / partner / sibling / friend / grandparent).

```
lib/band-roaming.ts
```


14 · Advanced features — Phase G · platform integrations

Phase G — Platform integrations

Make HearMe reachable from every surface the user uses.

28 Wear OS / watchOS companion

[ADAPTER + RECIPE](#)

Long-press a complication to fire SOS, double-tap for live-share, stream HR samples from the watch's PPG. The JS adapter accepts WatchPing events; the WatchConnectivity (iOS) and WearOS MessageClient (Android) recipes are bundled in the source.

`lib/wearable-companion.ts`

29 OS quick-action surface

[SHIPPED](#)

A deep-link endpoint (`hearme://sos`) that any OS surface can trigger: Android Quick Settings tile, iOS 17+ Action Button, iOS 18 Control Center widget, Siri shortcut donation. Adapter contract + native recipes bundled.

`lib/os-quicktile.ts`

30 CarPlay / Android Auto

[ADAPTER + RECIPE](#)

Lifecycle hooks (`notifyCarConnected` / `notifyCarDisconnected`) so JS code re-tunes voice trigger and journey monitor for in-car use. Big-button crash prompt template + post-fire confirmation banner. Swift CarPlay scene + Kotlin CarAppService recipes bundled.

`lib/auto-mode.ts`

31 Smart-home siren

[SHIPPED](#)

SOS fan-out to Hue (red flash via local bridge), Google Home / Nest (TTS via Home Assistant), Alexa Routine (webhook), and generic IFTTT / webhook receivers. All endpoint secrets live in SecureStore, not AsyncStorage.

`lib/smart-home.ts`

32 PSAP / RapidSOS adapter

[ADAPTER + RECIPE](#)

Forward an incident directly to the dispatch centre's API where regulation allows. RapidSOS sketch (OAuth + REST), EU 112 AML notes, and a tel:-fallback default. Real credentials are injected at registration time so they never enter the open-source build.

`lib/psap.ts`

15 · Advanced features — Phase H · operational

Phase H — Operational

Keep the system honest, observable, and self-healing.

33 Anonymous incident heatmap

SHIPPED

Each fired SOS appends a 3-decimal-place bucketed ping (~110 m grid). The public RPC `heatmap_at(lat, lon, radius)` aggregates per cell with a k-anonymity floor — cells with fewer than K hits collapse into a "rare" bucket. Users can erase their own contributions.

`lib/incident-heatmap.ts` · `supabase/heatmap.sql`

34 Self-test scheduler

SHIPPED

Nine non-destructive probes — SMS path, contacts validity, GPS accuracy, mic, biometric enrolment, NeuroBand link recency, Android battery optimisation hint, Supabase auth, device battery level. Runs weekly and on demand; surfaces broken pipes BEFORE an emergency.

`lib/self-test.ts`

35 Responder dashboard

SQL + ARCHITECTURE

Schema for partner organisations (NGOs, campus security): organisations, members, user-org links with explicit per-scope consent, and an immutable audit table. Two security-definer RPCs (`responder_active_alerts` , `responder_user_evidence`) enforce membership + consent + audit-on-read. Architecture spec lives at `docs/responder-dashboard.md` ; the dashboard itself is a separate Next.js project.

`supabase/responder.sql` · `docs/responder-dashboard.md`

16 · Permissions matrix

Permission	iOS	Android	Why HearMe needs it
Location (always & in-use)	✓ NSLocationAlways*	✓ ACCESS_FINE_LOCATION + COARSE	SOS coordinates, journey monitoring, off-grid watchdog, isolation score.
Camera	✓ NSCameraUsage	✓ CAMERA	Hidden-camera detector; evidence photos; QR scanning for witness mode.
Microphone	✓ NSMicrophoneUsage	✓ RECORD_AUDIO	Voice trigger, pre-roll buffer, evidence audio, acoustic events.
Contacts	✓ NSContactsUsage	✓ READ_CONTACTS	Pull trusted contacts during onboarding; never uploaded.
Face / Touch ID	✓ NSFaceIDUsage	✓ USE_BIOMETRIC + USE_FINGERPRINT	App lock, force-unlock detection, sensitive-screen gate.
Photo library	✓ NSPhoto*	✓ READ/WRITE_MEDIA	Save evidence photos; pick avatar.
Bluetooth (always & peripheral)	✓ NSBluetoothAlways*	✓ BLUETOOTH/SCAN/CONNECT	NeuroBand pairing, stalker scan, isolation neighbour count, community mesh.
SMS	composer-only	✓ SEND_SMS	Fire SOS to trusted contacts.
Phone	tel: scheme	✓ CALL_PHONE	Auto-dial emergency number; helplines.
Foreground service	—	✓ FOREGROUND_SERVICE + CONNECTED_DEVICE	Background BLE link to NeuroBand; journey-monitor location updates.
Vibrate	—	✓ VIBRATE	Band-feedback fallback patterns.
Background modes	✓ bluetooth-central	—	BLE central role active in background.

Each permission is requested with a clear rationale at the moment it's first needed. Permissions are minimised: e.g. journey monitoring is the only feature that asks for "always" location, and it asks only when the user starts a journey.

17 · Privacy & security model

Identity

- Multi-path auth (PIN, biometric, Aadhaar OTP, demo PIN). Losing one path doesn't lock the user out.
- Per-device Ed25519 signing key generated on first use, persisted in SecureStore (Keychain on iOS, EncryptedSharedPreferences on Android). Never leaves the device.

Data at rest

- **SecureStore:** duress PIN, signing private key, voiceprint enrolment, smart-home endpoint secrets, voice-trigger challenge, NeuroBand session key + PSK.
- **AsyncStorage:** contacts, settings, evidence sessions, alert history, NeuroBand baselines / event log, cycle log, dead-man queue, stalker sightings.
- **Supabase Storage (private bucket):** uploaded evidence files. Access only via signed URLs scoped to the owning user.

Data in transit

- **Live-share & witness mode:** AES-256-GCM with per-session keys. The key lives in the URL fragment (after #) — it never reaches a server.
- **Two-way duress & guardian relay acks:** Supabase realtime broadcasts. Payloads are E2E-encrypted where they carry sensitive data; ack signals carry only contact id + name.
- **Community mesh:** opt-in beacons carry only lat/lon/severity + a rotating opaque id. No user identity or device id.

Tamper evidence

- Every evidence item carries a SHA-256 content hash.
- Items are linked: `chain[i] = SHA256(chain[i-1] || contentHash[i])`. Any edit invalidates every later hash.
- The final chain hash is signed with the device Ed25519 key and submitted to a public OpenTimestamps Bitcoin calendar. Within ~6 hours the OTS receipt upgrades to a Bitcoin block attestation that anyone can verify offline.

Row-level security (Supabase)

- Every table that holds user data has RLS enabled. Policies restrict select / insert / update to `auth.uid() = user_id`.
- Storage policies are file-path-aware: a user can only write to `<their-uid>/...` paths in the evidence bucket.
- Public RPCs (`heatmap_at` , `evidence_attestation` , `responder_*`) are `security_definer` and enforce explicit consent / k-anonymity / role gating before returning rows.

Adversarial robustness

- Disguise mode (calculator) hides the app from a casual snoop.
- Decoy lock screen + duress PIN deflect an active attacker.
- Force-unlock detection escalates if struggle motion is detected during a bio prompt.

- Anti-uninstall guard (Android Device Admin) requires the duress PIN to remove the app.
- Phone-tamper + airplane-mode counters detect attempts to silence the phone.
- Deepfake voice-trigger guard requires a voiceprint match, demotes (never silently rejects) on failure.

18 · Setup & deployment

Prerequisites

- Node.js ≥ 20 and npm
- Expo CLI (use `npx expo`)
- Android Studio (emulator or device) and/or Xcode (macOS)
- A Supabase project (free tier is fine)
- (Optional) nRF Connect SDK for NeuroBand firmware

Install

```
git clone https://github.com/Debjyoti-sarkar/HearMe.git
cd HearMe
npm install
```

Configure environment

Create a `.env.local` file at the project root:

```
EXPO_PUBLIC_SUPABASE_URL=https://<project>.supabase.co
EXPO_PUBLIC_SUPABASE_KEY=<anon-key>
```

Use the anon key, not the service-role key. RLS gates every table.

Provision Supabase

Run the SQL files in order in the Supabase SQL editor:

1. `supabase/setup.sql` — base tables & profile RLS
2. `supabase/rls.sql` — additional policies
3. `supabase/evidence.sql` — evidence locker tables & bucket
4. `supabase/evidence-signatures.sql` — Ed25519 + OTS attestations
5. `supabase/community-mesh.sql` — community SOS beacons
6. `supabase/heatmap.sql` — k-anonymous incident heatmap
7. `supabase/responder.sql` — responder dashboard schema

Then enable Phone OTP under Auth → Providers.

Run on device

Several features (BLE, foreground services, biometrics) require a custom dev client — Expo Go isn't enough.

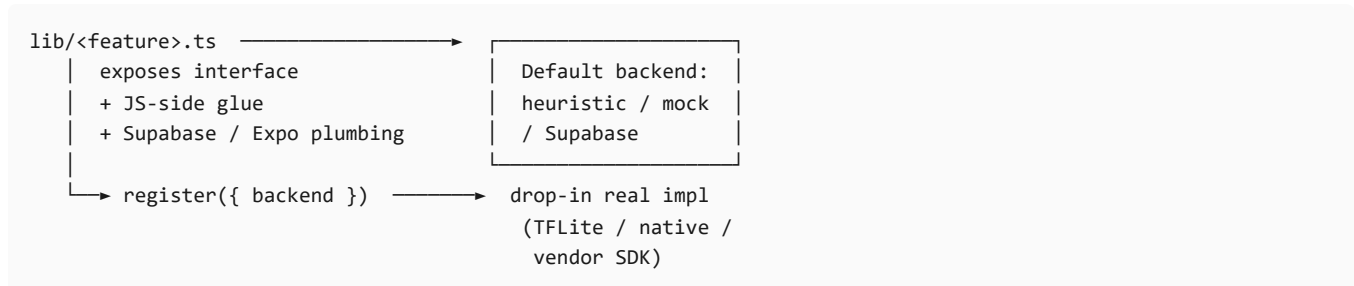
```
npx expo run:android      # adb-connected device
npx expo run:ios          # macOS + Xcode
# Distributables:
npx eas build --profile preview --platform android
npx eas build --profile production --platform ios
```

Flash NeuroBand firmware

```
cd firmware/neuroband
west build -b nrf52840dk/nrf52840 .
west flash
```

19 · Adapter pattern (production-grade strategy)

Several advanced features genuinely need code that is not in the JS / Expo surface — PSAP REST clients, Wear OS bridges, Android Device Admin, ML Kit face detection, on-device LLMs, satellite SDKs, Meshtastic protobufs. Rather than ship fake "production" code, each of those modules implements the **adapter pattern**:



The app calls `lib/<feature>` in one place. The default backend is a working, useful approximation — or, where there is no useful default (e.g. face detection), an honest no-op that *refuses to claim success it didn't earn*. Replacing it with the real thing is a single `register({ backend: realImpl })` call from a custom dev client.

Where the adapter pattern shows up

Module	Default backend	What plugs in
Pre-roll buffer	Chunk-rotating expo-av	AVAudioEngine / AudioRecord native circular buffer
Off-grid mode	Watchdog only — banners on loss	Meshtastic, Garmin inReach, sat-phone bridges
Anti-uninstall	No-op (reports unsupported)	Android Device Admin / iOS Screen Time
Threat classifier	Rule-based heuristic	On-device LLM (Gemma 2B / Phi-3-mini)
Acoustic events	Envelope detector	YAMNet TFLite
Diarization	RMS+ZCR+centroid voiceprint	pyannote / TitaNet / ECAPA-TDNN
Auto-redaction	Honest no-op (reports unsupported)	ML Kit (Android) / Vision (iOS)
Wearable companion	No-op	WatchConnectivity (iOS) / WearOS MessageClient
OS quick-tile	Deep-link only	TileService / Action Button bridge
CarPlay / Auto	Lifecycle no-op	CarPlay Scene / CarAppService
PSAP	tel: fallback	RapidSOS REST client / EU 112 AML

20 · File index — every module shipped

Domain logic — lib/

File	Purpose
lib/aadhaar.ts	Aadhaar e-KYC integration helpers.
lib/acoustic-events.ts	Acoustic event detection (envelope + YAMNet adapter).
lib/airplane-counter.ts	Airplane-mode toggle counter + alt-modem escalation.
lib/alert-history.ts	Chronological event log.
lib/anti-uninstall.ts	Anti-uninstall guard adapter (Android Device Admin recipe).
lib/app-data.ts	Contacts / settings AsyncStorage layer.
lib/audio-recorder.ts	Manual evidence recording flow.
lib/auto-mode.ts	CarPlay / Android Auto adapter.
lib/auto-redaction.ts	Face / plate / PII redaction adapter.
lib/band-feedback.ts	Extended band-side LED + haptic codes.
lib/band-roaming.ts	Group band roaming (multi-band per account).
lib/bba-model.ts	Behavioural Biometric Authentication model.
lib/behavior-detector.ts · behavior-tracker.ts	Behaviour anomaly detection engine.
lib/community-mesh.ts	Geo-anonymous community SOS mesh.
lib/cycle-aware.ts	Cycle-aware sensitivity (manual log + Health adapter).
lib/decoy-lock.ts	Fake lock screen with silent SOS on engage.
lib/demo-auth.ts	Offline demo PIN auth path.
lib/diarization.ts	Speaker diarization (voiceprint + adapter).
lib/drive-passenger.ts	Driver vs passenger classifier.
lib/duress.ts	Reversed / explicit duress PIN classification.
lib/emergency-sms.ts	SOS SMS pipeline + tel: dial.
lib/evidence-cloud.ts	Sync evidence sessions to Supabase.
lib/evidence-deadman.ts	Dead-man's switch retry queue.
lib/evidence-export.ts	Lawyer / NGO bundle (manifest + summary).
lib/evidence-locker.ts	Evidence-session data model + chain hashing.
lib/evidence-signing.ts	Ed25519 signing + OpenTimestamps anchor.

File	Purpose
<code>lib/force-unlock.ts</code>	Struggle-aware bio guard.
<code>lib/guardian-relay.ts</code>	Tier-staggered ack-aware fan-out.
<code>lib/hrv-baseline.ts</code>	Circadian HRV baselines + stress index.
<code>lib/i18n.ts</code>	i18next configuration.
<code>lib/incident-heatmap.ts</code>	K-anonymous heatmap reader / writer.
<code>lib/isolation-score.ts</code>	BLE + audio + light + speed isolation score.
<code>lib/journey-monitor.ts</code>	Route + deviation watcher.
<code>lib/live-share.ts</code>	E2E AES-GCM live-share over realtime.
<code>lib/nav-guide.ts</code> · <code>nav-guide-ref.ts</code>	Voice-guided navigation refs.
<code>lib/neuroband-ble.ts</code>	BLE bridge (real + mock).
<code>lib/neuroband-fusion.ts</code>	2-of-5 marker fusion engine (pure).
<code>lib/neuroband-pairing.ts</code>	Pairing flow with passkey + bonding.
<code>lib/neuroband-storage.ts</code>	Baselines + pair record + secure storage.
<code>lib/noble.d.ts</code>	Type shims for @noble crypto packages.
<code>lib/offgrid.ts</code>	Off-grid watchdog + transport adapter.
<code>lib/os-quicktile.ts</code>	OS quick-action surface adapter.
<code>lib/otp.ts</code>	OTP helpers.
<code>lib/phone.ts</code>	Phone-number normalisation.
<code>lib/phone-tamper.ts</code>	Phone-tamper triple-coincidence detector.
<code>lib/preroll-buffer.ts</code>	Continuous 30-second pre-roll audio buffer.
<code>lib/psap.ts</code>	PSAP / 112 / RapidSOS adapter.
<code>lib/rideshare.ts</code>	Rideshare-aware trip helpers.
<code>lib/safe-zones.ts</code>	Safe-zone geofences.
<code>lib/safety-score.ts</code>	Aggregate safety dashboard score.
<code>lib/self-test.ts</code>	Nine-probe weekly self-test.
<code>lib/session.ts</code>	Session lifecycle helpers.
<code>lib/siren.ts</code>	Loud-alarm playback.
<code>lib/smart-home.ts</code>	Hue / Google / Alexa / IFTTT fan-out.
<code>lib/stalker-scan.ts</code>	BLE stalker detector with persistent log.
<code>lib/supabase.js</code> · <code>supabase.d.ts</code>	Supabase client.
<code>lib/threat-classifier.ts</code>	Multi-modal threat score.

File	Purpose
lib/timer-checkin.ts	Check-in timer logic.
lib/trust-engine.ts	Trust-engine scoring.
lib/two-way-duress.ts	Contact challenge → band haptic.
lib/types.ts	Shared types (EmergencyContact, Settings, ...).
lib/voice-guide.ts	Spoken navigation engine.
lib/voice-trigger.ts	Voice trigger amplitude listener.
lib/voice-trigger-guard.ts	Deepfake voice-trigger verification.
lib/wearable-companion.ts	Wear OS / watchOS adapter.
lib/witness-mode.ts	QR-driven anonymous corroborating recorder.

Database — supabase/

File	Purpose
supabase/setup.sql	Profiles + emergency_logs + base RLS.
supabase/rls.sql	Additional row-level policies.
supabase/evidence.sql	Evidence locker tables + private bucket.
supabase/evidence-signatures.sql	Ed25519 + OTS attestations + public RPC.
supabase/community-mesh.sql	Mesh beacon + helper reply tables.
supabase/heatmap.sql	K-anonymous incident heatmap RPC.
supabase/responder.sql	Partner organisations + audit log + RPCs.

Documentation — docs/

File	Purpose
docs/ADVANCED_FEATURES.md	Master tracking doc for all 35 advanced modules.
docs/responder-dashboard.md	Responder dashboard architecture spec.
docs/HearMe_Complete_Guide.html	This guide — comprehensive overview.
docs/HearMe_Competitive_Analysis.*	Landscape of personal-safety apps.
docs/HearMe_Hardware_IoT_Strategy.*	Wearable roadmap, sensor stack.
docs/HearMe_NeuroBand_Integration.*	BLE protocol, fusion logic, security envelope.